

Given: Due:	Further Maths (Core Pure) HW 28	Name:
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Chapter 5: Volumes of Revolution

You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)
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Useful resources:

Your notes from recent lessons & your text book

Questions - do these on lined paper

Q1. The finite region S , shown shaded in Figure 2, is bounded by the y -axis, the x -axis, the line with equation $x = \ln 4$ and the curve with equation

$$y = e^x + 2e^{-x}, \quad x \geq 0$$

The region S is rotated through 2π radians about the x -axis.

Use integration to find the exact value of the volume of the solid generated.

Give your answer in its simplest form.

[Solutions based entirely on graphical or numerical methods are not acceptable.]

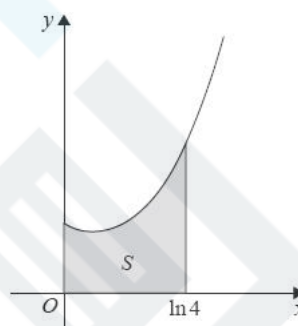


Diagram not drawn to scale

(7)

Q2.

$$f(x) = 2x^{\frac{1}{3}} + x^{-\frac{2}{3}} \quad x > 0$$

The finite region bounded by the curve $y = f(x)$, the line $x = \frac{1}{8}$, the x -axis and the line $x = 8$ is rotated through θ radians about the x -axis to form a solid of revolution.

Given that the volume of the solid formed is $\frac{461}{2}$ units cubed, use algebraic integration to find the angle θ through which the region is rotated.

(Total for question = 8 marks)

Q3.

Diagrams not drawn to scale.

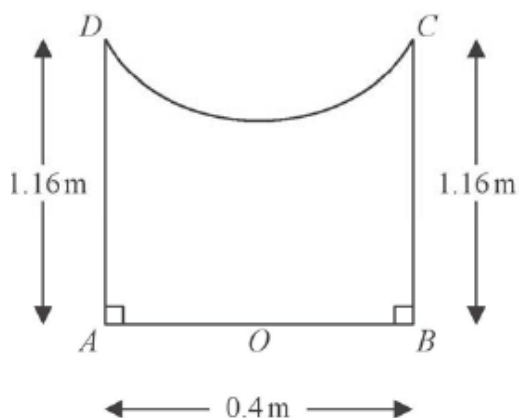


Figure 1

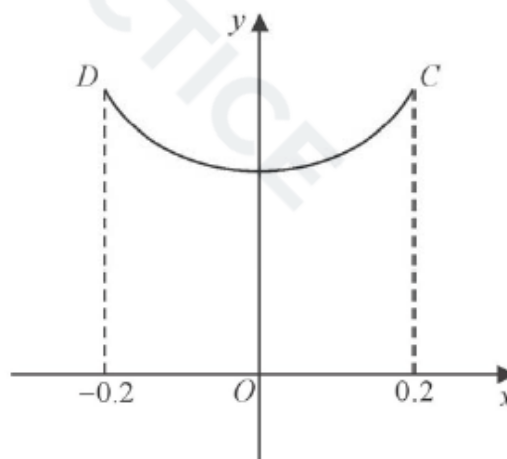


Figure 2

Figure 1 shows the central cross-section $AOBCD$ of a circular birdbath, which is made of concrete. Measurements of the height and diameter of the birdbath, and the depth of the bowl of the birdbath have been taken in order to estimate the amount of concrete that was required to make this birdbath.



Using these measurements, the cross-sectional curve CD , shown in Figure 2, is modelled as a curve with equation

$$y = 1 + kx^2, \quad -0.2 \leq x \leq 0.2,$$

where k is a constant and where O is the fixed origin.

The height of the bird bath measured 1.16 m and the diameter, AB , of the base of the birdbath measured 0.40 m, as shown in Figure 1.

- Suggest the maximum depth of the birdbath. (1)
 - Find the value of k . (2)
 - Hence find the volume of concrete that was required to make the birdbath according to this model. Give your answer, in m^3 , correct to 3 significant figures. (7)
 - State a limitation of the model. (1)
- It was later discovered that the volume of concrete used to make the birdbath was 0.127 m^3 correct to 3 significant figures.
- Using this information and the answer to part (c), evaluate the model, explaining your reasoning. (1)

Q4.

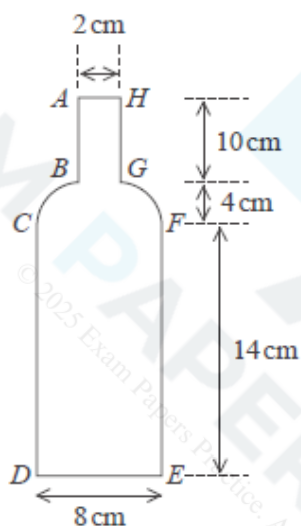


Figure 1

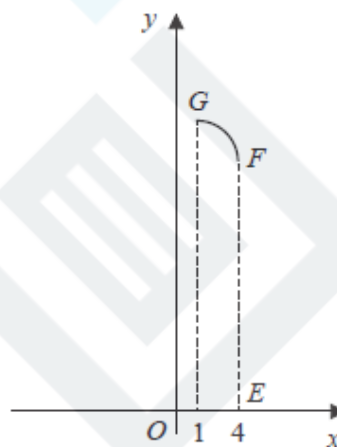


Figure 2

A mathematics student is modelling the profile of a glass bottle of water. Figure 1 shows a sketch of a central vertical cross-section $ABCDEFGHA$ of the bottle with the measurements taken by the student.

The horizontal cross-section between CF and DE is a circle of diameter 8 cm and the horizontal cross-section between BG and AH is a circle of diameter 2 cm.

The student thinks that the curve GF could be modelled as a curve with equation

$$y = ax^2 + b, \quad 1 \leq x \leq 4,$$

where a and b are constants and O is the fixed origin, as shown in Figure 2.

- Find the value of a and the value of b according to the model. (2)
 - Use the model to find the volume of water that the bottle can contain. (7)
 - State a limitation of the model. (1)
- The label on the bottle states that the bottle holds approximately 750 cm^3 of water.
- Use this information and your answer to part (b) to evaluate the model, explaining your reasoning. (1)

END OF HOMEWORK

Total = 38 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.